Two thresholds for the same four reports

The target Q is the supplied sixteen-word law. The target remains fixed while the null error allowance ε varies. With known ideal Bernoulli rate s=131/320, fresh at every prefix w, both directional error caps permit exactly $s(1-\varepsilon)\le q(w)\le s+(1-s)\varepsilon$, where q(w) is the next reported-one probability. Each allowed value is achievable by adjusting just the needed direction. Applying that construction at all fifteen nodes realizes the full law.

Observation retained	First compatible uniform cap
Odd parity alone, s fixed	4.41216575004%
All sixteen words, s fixed	11.28717824431%
All words, one common ideal rate freely fitted	8.15529322792%

For parity, let O be the odd probability. Starting at zero update O to $s+(1-s)\varepsilon+(1-2s)(1-\varepsilon)O$ four times to obtain the maximum. It first reaches the target odd probability 0.526209351956… at the listed cap. The admissible classes are nested. At threshold the maximizing rule matches the target parity, and the same rule remains admissible at larger caps.

For the full word, take the largest of the two directional requirements across positive-probability prefixes. The binding constraint is a false negative after 111: q(111)=0.363168114062… and the minimum cap is $(s-q(111))/s=10396360128757/92107698697800$. For this particular Q, an independently verified upper bound below that threshold on just this direction and prefix would already exclude exact agreement at the population level, provided s applies there. Finite records require uncertainty bounds and the prefix must occur.

If s is allowed to vary but remains common across every node, the admissible interval has width ε. The target therefore requires $\varepsilon\ge q_{max}-q_{min}$. Equality is attained with $s=q_{min}/(1-\varepsilon)=0.39541539935\ldots$. The minimum and maximum target rates are 0.36316811406… and 0.44472104634…. This sensitivity check changes the identified quantity’s calibration assumption, using the same target table.

At a 1% cap, the fixed-prefix minimum total-variation distance is 0.021712152209. Allowing the prefix to vary subject to all four per-position directional caps gives 0.020642673036. The earlier exact constructive and dual certificates and the submitted replay agree. These are distances between distributions. They are distinct from both the cap thresholds and the mismatch percentages discussed in Freshness versus Reporting Error.

What the zero-error counts buy

Assume fixed sample sizes, correctly labelled independent Bernoulli observations within each of 30 cells, fixed rates and valid transfer to the gate path. A cell is defined by position, prior reported history and ideal bit value. No independence between cells is needed for the two guarantees below.

Calibration objective at 99% coverage	Zero errors in each cell	Total cell observations
Thirty separately reported limits using equal Bonferroni allocations	797 per cell	23,910
Single upper limit on the largest of all thirty rates	459 per cell	13,770

For n zero-error trials, the one-sided exact binomial upper limit at failure probability γ is $1-\gamma^{(1/n)}$. Allocating γ=0.01/30 to each cell yields 797. For the common cap, compute a one-sided 99% limit U_i for each rate p_i and report $U_{max}=\max U_i$. For a fixed cell i attaining the true maximum, $U_{max} \geq U_i$, so it covers the maximum with probability at least 99%. This gives one valid common bound for all cells. It does not confer simultaneous coverage on thirty individually tighter limits.

Equivalently, if some cell has rate at least 1%, the chance all cells have zero errors is at most 0.99^n. This quantity is 0.01002119 at n=458 and 0.00992097 at n=459. Other cells can have zero true rate, so a smaller fixed equal sample size cannot certify the same pass-on-all-zeros rule at 99%. Cell observations are not counts of complete records.

Held-out power and the final decision

At 5% size the parity-only power is 0.949970847338 at 6,342 and 0.950004432130 at 6,343. At 4% it is 0.949974789382 at 6,757 and 0.950018042463 at 6,758. Exact integer binomial tails verify the adjacent crossings with randomized decision at the cutoff; an extra record can always be discarded.

The proposed 1%+4% allocation is sufficient. A different fixed rule issues a rejection only when calibration certifies the prespecified cap AND the gate test rejects. If the actual common cap fails, false joint rejection is bounded by γ; if it holds under the null, it is bounded by the gate-test size α. Thus the unconditional size is at most max(γ,α). Keeping γ=1% and α=5% permits the 6,343 held-out crossing. This argument needs stable transferable rates and a fixed decision rule. A 95%-powered held-out test supplies no 95% guarantee that calibration passes and the whole procedure rejects.

Exact binomial limits, NIST; see also Berger & Hsu on intersection–union tests.